"Stock markets are efficient. Therefore, investors should simply follow a passive strategy… Active fund management is wasteful and earns inferior returns!"
Discuss this statement, with particular reference to behavioural theories.

Introduction

The concept of market efficiency is derived from the "fair game" model, which evolved from the "random walk" theory meaning that successive price changes are statistically independent and unpredictable. Most financial economists are of the opinion that this framework became the foundational pillar of modern economics. The term is used to refer to a market in which prices "fully reflect" all available information (Fama, 1970). This hypothesis is typically categorized into weak, semi-strong, and strong forms, depending on whether the information set includes past prices, public announcements, or monopolistic data. Ultimately, the modern rise of index funds reflects a widespread commitment to these principles, as the theory suggests that passive strategies are generally superior to costly active fund management.

To understand the practical implications of market efficiency, it is necessary to examine how the efficient markets theory translates into investment strategy. Passive investing is an investment strategy defined as an approach in which an investor seeks to hold a "market" portfolio rather than attempting to beat the market through active selection. In this framework, it is assumed that security prices "fully reflect" all available information, thereby acting as accurate signals for resource allocation (Fama, 1970). Therefore, in an efficient market, active fund management is viewed as wasteful because the significant costs of research and trading frequently lead to returns that are inferior to a simple buy-and-hold position.

Moreover, in behavioural finance, market participants are frequently considered to be not fully rational when processing new information or making investment choices (Barberis and Thaler, 2002). Behavioural finance can be understood as a field that combines insights from psychology with the idea of limits to arbitrage in order to explain why asset prices can move away from their fundamental values.

The objective of this essay is to show the nature and logic of Efficient Capital Markets as defined by Fama (1970). The central argument is that while the Efficient Market Hypothesis has shaped modern practice, behavioural finance provides evidence of persistent inefficiencies. Consequently, the dominance of passive investing may reflect practical constraints rather than the total absence of exploitable mispricing.

Efficient Market Hypothesis

Theoretical characteristics of efficient capital markets refer to the conditions where security prices provide accurate signals for resource allocation. It includes the full reflection of available information, the rapid adjustment to new data, and the absence of systematic patterns. Information is deemed to be efficient if it is fully utilized in the process of price formation. According to Fama (1970), a market is efficient when security prices "fully reflect" all available information, meaning that the current price a stock is trading for today is its fair value (Ricciardi and Simon, 2000). This implies that information must be used so effectively that no participant

can consistently achieve economic profits, that is, risk-adjusted returns after all costs, by trading on any particular set of information.

Another key concept, will be the categories of market efficiency refer to the specific information sets that are incorporated into security prices. Categories of market efficiency include weak form, semi-strong form, and strong form. Weak form efficiency entails the belief that the information set consists solely of the historical sequence of prices or returns, suggesting that past patterns cannot predict future performance (Jensen, 1978). Semi-strong form efficiency entails the capacity of the market to immediately incorporate all obviously publicly available information, such as earnings announcements, stock splits, or new security issues. Strong form efficiency involves the reflection of all relevant information, including private data known only to corporate insiders or specialists with monopolistic access.

Concepts of stochastic price movements refer to the mathematical frameworks used to describe unpredictable changes in security values (Fama, 1970). Concepts of stochastic price movements include the random walk, the "fair game" model, and the Martingale property (Samuelson, 1965). The random walk model essentially involves the statistical independence of successive price changes, where the past cannot be used to increase expected gains beyond a naive buy-and-hold strategy. A price series therefore is considered a "fair game" if there is no way of making an expected profit by extrapolating past changes in quotations. In a Martingale model, properly anticipated prices fluctuate randomly because the market has already discounted all knowable future contingencies as much as is humanly possible.

The existence of market efficiency is mainly determined by the speed and accuracy of information flow. Such information efficiency involves the importance of information being available to all market participants. An item consequently is considered to be incorporated into prices if "sufficient numbers" of investors have ready access to it and can make unbiased evaluations. According to Fama (1979), a market can be efficient if there are no transaction costs, information is freely available, and all participants agree on the implications of current data. However, these are sufficient rather than necessary conditions, as markets can still be efficient even in the presence of transaction costs or disagreement, provided no investor can consistently make better evaluations than those implicit in market prices.

If markets process information in this manner, the implications extend beyond theory and directly influence how rational investors should structure their investment strategies. Rational strategies for investors refer to the methodologies adopted by participants in response to the foundational belief that markets are efficient. Rational strategies for investors include passive fund management, the holding of broad-based index funds, and the implementation of a long-term "buy and hold" strategy (Ricciardi and Simon, 2000). Passive investing is an essential characteristic of modern practice, and it implies that investors should own the "entire market" rather than attempting to beat it through active selection. In this regard, active fund management is viewed as wasteful because the significant costs of research and trading, frequently lead to returns that are inferior to a simple market-tracking position. Understandability and usefulness of passive strategies go together, as data indicates that broad

stock indexes like the S&P 500 outperform most active managers over long periods (Ricciardi and Simon, 2000).

The logic of passive investing with regard to decision makers is dependent on the recognition that stock markets are a "fair game". However, the commitment to this belief depends mainly on the empirical evidence showing that professional analysts are often unable to forecast returns accurately enough to recover their costs. Supporters of the Efficient Market Hypothesis argue that since stocks are at their fair value, active traders cannot produce superior returns over time. Hence, the modern rise of index funds reflects the premise that because information is utilized so effectively, most investors are best served by minimizing transaction costs and accepting the market rate of return. This strategy assists investors in avoiding mental mistakes and emotional impulses that often lead to excessive trading and inferior outcomes.

Evidence Supporting Passive Investing

Data-based evidence on passive investing supports the buy-and-hold approach rather than active management, while empirical characteristics of passive investing support include the underperformance of active funds, the negative impact of high turnover, the significance of management costs, and the distortion caused by survivorship bias (Carhart, 1997). Performance is deemed to be supportive of passive strategies if it is capable of influencing an investor's decision to avoid active security selection (Ricciardi and Simon, 2000). According to standard financial research, investment outcomes should be analysed net of all costs and adjusted for risk for them to be regarded as valid evidence (Jensen, 1978).

The underperformance of professionally managed funds is mostly determined by their inability to recover research and transaction costs (Fama, 1970). Such professional underperformance involves the failure of the majority of active managers to produce returns that beat a comparable market index. A fund consequently is considered inferior if its risk-adjusted returns net of all costs are lower than those achieved by a randomly selected portfolio or broad-based index. A study indicates that approximately 60% to 80% of the time, broad market indexes like the S&P 500 outperform the universe of active equity funds (Ricciardi and Simon, 2000).

Regarding trading frequency, a strategy is considered efficient if it minimizes unnecessary transaction volume and represents faithfully the benefits of long-term holding. According to behavioural and standard finance research, efficient trading must be able to meet conditions where turnover remains low so that commissions do not erode investor capital (Ricciardi and Simon, 2000). This reinforces the idea that a long-term buy-and-hold strategy is required to avoid the mental mistakes and emotional impulses that lead to excessive trading. Barberis and Thaler (2002) confirm that the most active traders in their sample earned the lowest average returns specifically because high turnover increases the weight of transaction expenses.

The importance of investment costs is determined by their long-term impact on the final wealth of the market participants (Carhart, 1997). These costs are management fees, administrative expenses, 12b-1 charges, and brokerage commissions. Jensen (1968)

highlights that even well-endowed mutual fund analysts are unable to forecast returns accurately enough to recover these research and transaction costs. Therefore, of the annual spread in mean return between top and bottom funds, differences in expense ratios and transaction costs explain a significant portion of the measurable alpha spread (Carhart, 1997).

With regard to survivorship bias, performance data is considered reliable only if it includes all funds that were part of the opportunity set at the start of a period (Edwards et al., 2025). Survivorship bias involves the tendency of databases to include only the survivors, thereby missing those funds that were liquidated or merged out of existence during the study. This indicates that the records of surviving funds overstate the success of active management by ignoring the long-run losers. Consequently, a sample free of survivor bias demonstrates that common factors and investment expenses almost completely explain the predictability in mutual fund returns (Carhart, 1997).

The usefulness of passive strategies for decision makers depends on understanding that stock markets work as a kind of "fair game." This means the information in financial statements should be used to reduce costs, instead of trying to predict future results. From this point of view, investors recognize that high portfolio turnover and large fees usually lead to worse results than simply following the market. In the end, choosing a passive strategy mostly comes from accepting that, once management fees are included, most active funds do not manage to generate higher risk-adjusted returns than the market (Morningstar Manager Research et al., 2025).

Behavioural Challenges to Market Efficiency

While the Efficient Market Hypothesis (EMH) assumes rational investors and fast information processing, behavioural finance suggests that systematic cognitive biases and institutional constraints may prevent prices from always reflecting fundamental values (Ricciardi and Simon, 2000). This perspective provides a theoretical and a practical framework to understand why markets may not be perfectly efficient even if active fund management remains difficult to execute profitably (Barberis and Thaler, 2002).

A) Investor Irrationality

The psychological foundations of behavioural finance posit that if investors are systematically biased, security prices will inevitably deviate from their fundamental values. One prominent bias is loss aversion, derived from prospect theory, where investors assign more significance to losses than they do to equivalent gains (Barberis and Thaler, 2002). This manifests as the disposition effect, where investors hold losing stocks too long in the hope of breaking even, driven by the convexity of the value function in the domain of losses (Barberis, 2012).

Moreover, framing effects show that investors make different decisions based on how information is presented to them. For instance, investors evaluate risk differently if they view their portfolio through an annual rather than a monthly frame. Additionally, investors rely on heuristics, such as the law of small numbers, which causes them to overreact to short-term trends (Barberis and Thaler, 2002). If investors systematically misinterpret information through these psychological lenses, market prices reflect collective psychology rather than

purely fundamental value. These behavioural patterns surely challenge the EMH assumption that investors process information rationally, suggesting that price formation may be influenced by cognitive biases rather than informational efficiency.

B) Market Anomalies

In an efficient market, predictable return patterns should not exist because any available information should already be reflected in the price (Malkiel, 2003). However, research has identified several "anomalies" that contradict this premise (Carhart, 1997). For example, momentum effects show that stocks that have performed well in the recent short-term (6–12 months) tend to continue outperforming, which directly contradicts weak-form efficienc (Barberis, Mukherjee and Wang, 2014) y. Conversely, the value effect shows that stocks with high book-to-market ratios (value stocks) consistently earn higher average returns than growth stocks, challenging semi-strong efficiency.

Moreover, evidence of long-term investor overreaction shows that extreme past losers significantly outperform past winners over three-to-five-year horizons (DeBondt and Thaler, 1987). These anomalies imply predictable excess returns, which should not exist under a strict interpretation of the EMH. However, the interpretation of these findings remains a point of discussion as some proponents of EFM argue that these anomalies do not represent inefficiencies but rather are proxies for risk premia that are not captured by the simple CAPM.

C) Why Inefficiencies Persist

The most critical analytical component of the behavioural challenge is the theory of limits to arbitrage (Barberis and Thaler, 2002). Without this building block, the existence of irrational investors would be irrelevant, as rational "arbitrageurs" would quickly correct any mispricing (Barberis and Thaler, 2002). However, behavioural finance argues that correcting mispricing is often too risky and costly to be effective. Arbitrageurs face fundamental risk because substitute securities are rarely perfect, leaving them vulnerable to firm-specific news (Barberis and Thaler, 2002). Even more significant is noise trader risk, the danger that a mispricing worsens in the short run as irrational sentiment becomes even more extreme.

Institutional constraints further limit the ability to exploit inefficiencies. For example, short-sale constraints and high borrowing costs can prevent traders from betting against overvalued stocks, as seen in the "Palm/3Com" carve-out case (Barberis and Thaler, 2002). Additionally, because many arbitrageurs manage other people’s money, a temporary widening of mispricing can lead to fund withdrawals, forcing them to liquidate positions at a loss. This explains why substantial inefficiencies can persist in the market even while active managers fail to beat the market, as the risks and costs of the required arbitrage strategies often outweigh the potential gains.

We can say that behavioural finance questions the strong assumptions behind market efficiency. It suggests that even though passive investing often performs better than most active strategies, this does not automatically mean markets are perfectly efficient. Instead, the

success of passive strategies may just show how difficult and risky it is to consistently take advantage of mispricing in the market, especially when cognitive biases and limits to arbitrage make these opportunities hard to exploit.

Is Active Management Necessarily Wasteful?

Even though the evidence shows that most active managers underperform market benchmarks after fees, this does not automatically mean that active management itself is wasteful (Jensen, 1968).

Managerial skill refers to the ability of a small number of professionals to generate persistent abnormal returns through better security selection (stock picking). While most funds fail to beat the market, research by Carhart (1997) shows that top-ranked funds often recover their costs, suggesting that skill does exist. Similarly, it can be argued that skill is present but often hidden by market equilibrium, where competition for capital and higher fees push observable alpha for investors close to zero. This signifies active management may not be wasteful in theory, but genuine skill is scarce and very difficult to identify beforehand.

Market efficiency also changes across segments, meaning the value of active management is not the same everywhere. It may be more useful in areas such as small-cap stocks, illiquid assets, or emerging markets, where information is weaker and analyst coverage is lower (Harvey and Liu, 2010). Data from Morningstar Manager Research et al. (2025) supports this, showing that active funds in emerging markets or small-growth categories often have higher success rates than those focused on large developed markets. This suggests that the dominance of passive investing may reflect the high efficiency of large markets rather than a universal advantage across all asset classes.

Finally, from an adaptive market perspective, efficiency is not fixed but evolves over time. According to this view, inefficiencies appear and disappear as investors compete, regulations change, and technology develops. As noted by Burton Malkiel (2003), predictable anomalies like the “January effect” tend to vanish once they become widely known and exploited. In this sense active management plays an important role in price discovery. This means that the effort to find mispricing is what eventually makes markets more efficient. For this reason, active management can still be valuable in changing or less stable environments, even if passive strategies dominate in mature markets.

Conclusion

Overall, the Efficient Market Hypothesis, linked to Eugene Fama (1970), remains very influential because it helps explain why passive investing tends to work well in practice. Evidence on fees, turnover, and fund performance shows that for most investors, trying to consistently beat the market is expensive and rarely successful. In that sense, markets seem efficient enough for passive investing to be the most rational default strategy.

However, insights from behavioural finance, particularly those associated to researchers such as Barberis (2012), make it almost impossible to accept that markets are perfectly efficient. Psychological biases, momentum patterns, and limits to arbitrage suggest that prices are

influenced not only by information, but also by human behaviour and institutional constraints. This means inefficiencies probably exist, even if exploiting them consistently is very difficult.

From my perspective, the most convincing practical solution is to combine a broadly diversified passive foundation with a more focused allocation to active strategies. Maintaining a primarily passive base lets investors capture overall market efficiency while keeping costs low. At the same time, dedicating a smaller portion of the portfolio to selective active decisions acknowledges that behavioural mispricing and rigorous research can create opportunities. In this framework, active management becomes a complementary tool rather than an unnecessary expense. It supports price discovery and can add value in specific situations.

Therefore, rather than viewing passive and active investing as opposites, but as mutually reinforcing. Markets may not be perfectly efficient, but they are efficient enough that a balanced strategy, combining passive discipline with selective active judgement, offers the most convincing approach.

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